The full low-temperature Ising susceptibility: a regular-coordinate proof of the natural boundary

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Abstract

We prove that the complete connected bulk susceptibility of the isotropic square-lattice Ising model, continued from its low-temperature pure phase, has the unit circle as a natural boundary in $s = \sinh(2\beta_{\mathrm{phys}}J)$. An exact lattice Fourier sum expresses each full coefficient as an angular integral with a total-angle constraint and one product denominator. At a dense family of prime roots of unity, the first singular even-particle term has a nonzero $\varepsilon^{-1/2}$ derivative, computed directly on this constraint. An elementary four-point determinant identity, propagated by An elementary four-point determinant identity, extended by Pfaffian condensation, gives the exact local coefficient. A deformation that preserves the total angle and a regular coordinate on the one-sided contour give an absolute logarithmic bound for the differentiated higher-particle tail. A single tangential field and oblique coarea control every right-side collision diameter, including diameters below the radial distance. A bounded anchor field treats the regions with a separated product kernel. These estimates also control the finite-order complement, so the first singularity survives in the full series.

1 Introduction

Consider the zero-field Ising model on the infinite square lattice, with equal positive nearest-neighbor coupling J. Set

$$s = \sinh(2\beta_{\mathrm{phys}}J), \qquad \mathcal{M}^2(s) = (1 - s^{-4})^{1/4},$$

where the branch of $\mathcal{M}^2$ tends to one at infinity. For real $s > 1$ the connected response in the plus pure phase is

$$\mathcal{X}(s) = \beta_{\mathrm{phys}}^{-1}\chi(s) = \sum_{(a,b)\in\mathbb{Z}^2} (\langle\sigma_{00}\sigma_{ab}\rangle_+ - \mathcal{M}^2(s)). \tag{1.1}$$

Spin reversal gives the same function in the minus pure phase. The physical inputs are Yang's magnetization formula and the low-temperature correlation expansion of Palmer–Tracy [1, 2], in the normalization recorded by Tracy–Widom [3, Appendix 1, pp. 1132–1133].

Nickel identified the unit-circle singularities of individual form factors [4, 5]. Their increasing density led to the natural-boundary conjecture for the full susceptibility; the series and amplitude analysis of Orrick, Nickel, Guttmann and Perk [6] provides further evidence. Passing from the singularities of individual terms to the infinite sum requires estimates that exclude cancellation. Tracy and Widom established a natural boundary for the diagonal susceptibility [7]. The full lattice sum in (1.1) requires control of both lattice directions.

The proof below establishes the natural boundary of the complete low-temperature bulk susceptibility using its constrained angular representation. A regular coordinate on the dispersion curve and a collective tangential field give a logarithmic bound for the complete differentiated tail.

Theorem 1.1. *The exterior holomorphic function $\mathcal{X}$ determined by (1.1) has the entire circle $|s| = 1$ as a natural boundary. The same local noncontinuation holds for the physical susceptibility $\chi = \beta_{\mathrm{phys}}\mathcal{X}$, with $\beta_{\mathrm{phys}}(s) = \operatorname{arsinh}(s)/(2J)$ continued from the real low-temperature interval.*

We use coefficients C_N that already contain the full lattice sum:

$$\mathcal{X} = \mathcal{M}^2 \sum_{\substack{N\geq 2\\ N \text{ even}}} C_N.$$

At a prime root $\zeta = e^{2\pi i a/p}$ in the first quadrant, the first singular even order is $N_* = 2p$. For $k = N_*^2/2 - 1$ and $s_\varepsilon = (1+\varepsilon)\zeta$, we prove

$$\begin{aligned} C_{N_*}^{(k)}(s_\varepsilon) &= \Lambda_\alpha \varepsilon^{-1/2} + o(\varepsilon^{-1/2}), \qquad \Lambda_\alpha \neq 0, \\ \sum_{\substack{N>N_*\\ N \text{ even}}} |C_N^{(k)}(s_\varepsilon)| &= O(1 + |\log \varepsilon|). \end{aligned} \tag{1.2}$$

All constants may depend on this fixed point and derivative order. The first estimate follows by rescaling the local shape variables. For the second, we deform the constrained angular torus and then partition it. Centering the regular coordinate at the moving branch point gives uniform coefficients for parameter differentiation. A tangential field preserves the angular constraint and total phase on the right part, while a bounded field anchored at a compact left root handles the part with a separated product kernel. Determinant minors give factorial decay, and coarea controls the dependence on the collision diameter. A fixed gap separates the finitely many small-order equality means from the branch.

AI Declaration

ChatGPT was used extensively in the mathematical research, proof development, and preparation of this manuscript.

2 The full angular representation

Write

$$S(s) = s + s^{-1}, \qquad W(y;s) = S(s) - \tfrac{1}{2}(y + y^{-1}), \qquad z = W - \sqrt{W^2 - 1} = e^{-\gamma}.$$

For $|s| > 1$ and $|y| = 1$, choose the unique interior root $|z| < 1$, continued from real $s > 1$. Indeed $S(s) \notin [-2, 2]$ in the exterior, so W cannot lie in $[-1, 1]$. The local logarithms γ have positive real part, and $\sinh\gamma = (z^{-1} - z)/2$ is single-valued.

On $y_i = e^{i\theta_i}$ put $Z = \prod_i z_i$ and

$$K_{il} = \frac{\sin((\theta_i - \theta_l)/2)}{\sinh((\gamma_i + \gamma_l)/2)}, \qquad K_{ii} = 0. \tag{2.1}$$

Changing a half-angle or logarithmic sign changes one row and its corresponding column, so the determinant is unambiguous and periodic. Each kernel entry has the rational square

$$K_{il}^2 = -\frac{(y_i - y_l)^2 z_i z_l}{y_i y_l (1 - z_i z_l)^2}. \tag{2.2}$$

The physical expansion uses the determinant directly. Section 4 gives its collision multiplicities and local coefficient.

Let the constrained torus and its normalized Haar measure be

$$\mathbb{H}_N = \{\theta \in \mathbb{T}^N : \textstyle\sum_i \theta_i = 0 \pmod{2\pi}\},$$

$$d\mu_N = \prod_{i=1}^{N-1} \frac{d\theta_i}{2\pi}, \qquad \theta_N = -\sum_{i<N} \theta_i \pmod{2\pi}.$$

Equivalently, $d\mu_N$ is Euclidean hypersurface measure divided by $\sqrt{N}(2\pi)^{N-1}$.

Proposition 2.1 (Complete lattice sum)**.** *For even $N \geq 2$, define*

$$C_N(s) = \frac{1}{N!} \int_{\mathbb{H}_N} \frac{1+Z}{1-Z} \det K \prod_{i=1}^{N} \frac{1}{\sinh \gamma_i} \, d\mu_N. \tag{2.3}$$

Then the exterior continuation of the physical susceptibility is

$$\mathcal{X}(s) = \mathcal{M}^2(s) \sum_{\substack{N \geq 2 \\ N \text{ even}}} C_N(s). \tag{2.4}$$

Every fixed differentiated series converges normally on compact exterior parameter sets. All lattice sites, including the origin, are included in (2.4)*.*

Proof. For real $s > 1$, the physical input is the Fredholm formula

$$\langle \sigma_{00} \sigma_{ab} \rangle_+ = \mathcal{M}^2 \det(I + g_{a,|b|}), \qquad g_{a,|b|}(\theta_i, \theta_l) = e^{ia\theta_i - |b|\gamma_i} K_{il},$$

with measure $d\theta/(2\pi \sinh \gamma)$. The originally stated kernel is $\sinh((\gamma_i - \gamma_l)/2)/\sin((\theta_i + \theta_l)/2)$. Subtracting the dispersion equations and using the difference formulas for cosine and hyperbolic cosine gives exactly (2.1), with removable values filled in by continuity. Each Fredholm minor is the product of its row weights and $\det K$. The odd minors vanish because K is skew. The expansion therefore gives

$$\langle \sigma_{00} \sigma_{ab} \rangle_+ = \mathcal{M}^2 \left(1 + \sum_{\substack{N \geq 2 \\ N \text{ even}}} c_N(a, b; s) \right),$$

where

$$c_N(a, b; s) = \frac{1}{N!} \int_{\mathbb{T}^N} e^{ia \sum_i \theta_i} Z^{|b|} \det K \prod_i \frac{d\theta_i}{2\pi \sinh \gamma_i}.$$

For this fixed s, all entries and one-body factors are bounded on a common small angular strip, on which $|z_i| \leq e^{-c} < 1$. Shift every angle by $i\eta \operatorname{sgn}(a)$, with fixed small $\eta > 0$. Periodicity and analyticity give the sitewise bound

$$|c_N(a, b; s)| \leq \frac{C^N N^{N/2}}{N!} e^{-N(\eta|a| + c|b|)}.$$

Hadamard's inequality supplies the determinant factor. The bound is summable in all sites and all even $N \geq 2$, so the lattice and particle sums may be interchanged.

Summing over $b \in \mathbb{Z}$ gives $(1+Z)/(1-Z)$. For the other lattice direction, associate to each periodic F on $\mathbb{T}^N$ a function of the total angle given by

$$Q(v) = \int_{\mathbb{T}^{N-1}} F(\theta_1, \ldots, \theta_{N-1}, v - \textstyle\sum_{i<N} \theta_i) \prod_{i<N} \frac{d\theta_i}{2\pi}.$$

Its Fourier coefficient of index $-a$ is $\int e^{ia\sum\theta_i} F \prod_i d\theta_i/(2\pi)$. For the present analytic integrand the Fourier series is absolutely convergent. Summing those coefficients gives $Q(0)$, precisely (2.3). Subtracting $\mathcal{M}^2$ in the correlation formula removes its zeroth-order term and proves (2.4). At the origin the full correlation is one, so its connected contribution in this sum is exactly $1-\mathcal{M}^2$.

On a compact exterior neighborhood the unit-circle roots satisfy $m \le |z_i| \le q < 1$. The complete kernel entries, one-body factors, and product denominator in (2.3) are uniformly bounded. Since $\mu_N(\mathbb{H}_N) = 1$, Hadamard and $N! \ge (N/e)^N$ give

$$|C_N(s)| \le C_0(C_1/\sqrt{N})^N.$$

Cauchy's inequality on a slightly larger compact neighborhood gives the same summability for every fixed derivative. Analytic continuation from the real interval proves the exterior identity. □

Changing s, y_i, z_i to $-s, -y_i, -z_i$ preserves the constrained torus when N is even. It changes each complete matrix entry and each one-body factor by a minus sign; the full coefficient is unchanged. Thus C_N and $\mathcal{X}$ are even. Their values also commute with complex conjugation, first on the physical germ and then by analytic continuation.

3 Prime roots with a unique first cosine pair

Choosing the two dispersion angles equal gives a dense family of roots of unity. At the first even singular order there is a unique cosine pair. Its two signed angular configurations are exchanged by reflection.

Lemma 3.1. *Let $p \ge 5$ be prime and $1 \le a < p/4$. Put*

$$\alpha = \frac{2\pi a}{p}, \qquad \zeta = e^{i\alpha}, \qquad S_* = 2\cos\alpha, \qquad N_* = 2p.$$

Then the first positive even integer n for which

$$S_* = \cos(2\pi l/n) + \cos(2\pi m/n)$$

is $n = N_*$. At that order the unordered cosine pair is uniquely $\{\cos\alpha, \cos\alpha\}$. These points ζ are dense in the open upper-right quadrant of the unit circle. If

$$b = \arccos(S_* - 1), \qquad \xi = e^{-ib},$$

then ξ is algebraic and is not a root of unity.

Proof. Write $\zeta_p = e^{2\pi i/p}$, $d = (p-1)/2$, and $u_j = \zeta_p^j + \zeta_p^{-j}$ for $1 \le j \le d$. In $E = \mathbb{Q}(\zeta_p + \zeta_p^{-1})$, every u_j has trace -1: its conjugates sum the nontrivial p-th roots of unity once each. In particular $S_* = u_a > 0$ is irrational, since rationality would give $du_a = -1$.

If $n < 2p$ is even, then $p \nmid n$. The compositum of $\mathbb{Q}(\zeta_p)$ and $\mathbb{Q}(\zeta_n)$ is $\mathbb{Q}(\zeta_{pn})$. Its cyclotomic degree is $\varphi(pn) = \varphi(p)\varphi(n)$, so the two fields have intersection $\mathbb{Q}$. Thus a Nickel equation of order n would make S_* rational. The displayed pair (α, α) is available at order $2p$, which is therefore the first even order.

Next $\xi + \xi^{-1} = 2u_a - 2$, so ξ is algebraic. If ξ were a root of unity, every conjugate of $2u_a - 2$ would lie in $[-2, 2]$. Its trace in E, however, is $-2 - 2d < -2d$. This proves nonresonance.

At order $2p$ a cosine equal to -1 is impossible, since its partner would have to be $S_* + 1 > 1$. A cosine equal to 1 would make ξ a $2p$-th root of unity. All remaining twice-cosines belong to $\{\pm u_1, \dots, \pm u_d\}$. The numbers $u_1, \dots, u_d$ are linearly independent over $\mathbb{Q}$: a relation yields

a polynomial of degree at most $p-1$, with zero constant coefficient, vanishing at ζ_p. It is a multiple of $\Phi_p(X) = 1 + X + \cdots + X^{p-1}$, and its constant coefficient forces that multiple to be zero. Consequently $2u_a = v + w$, with $v, w \in \{\pm u_j\}$, implies $v = w = u_a$.

Finally, given $\alpha_0 \in (0, \pi/2)$, choose unbounded primes p and integers a nearest to $p\alpha_0/(2\pi)$. Eventually $1 \le a < p/4$, and $2\pi a/p \to \alpha_0$. □

4 A product identity for the correlation determinant

A Vandermonde determinant gives the four-point identity on the dispersion curve. Pfaffian condensation then extends the identity to every even order, determining the signs and collision multiplicities.

Lemma 4.1 (Four points on the dispersion curve)**.** *Let nonzero y_i, z_i satisfy $y_i + y_i^{-1} + z_i + z_i^{-1} = 2S$ for $1 \le i \le 4$, and set $B_{ij} = (y_i - y_j)/(1 - z_i z_j)$. Then*

$$B_{12}B_{34} - B_{13}B_{24} + B_{14}B_{23} = \left(\prod_{i=1}^{4} \frac{z_i}{y_i}\right) \prod_{i<j} B_{ij}.$$

The equality holds wherever the expressions are defined and, after clearing denominators, at every point of the curve.

Proof. Write $a_{ij} = y_i - y_j$, $t_i = z_i + z_i^{-1}$, and $Z_4 = z_1 z_2 z_3 z_4$. For a partition $ij \mid kl$ of the four indices, direct multiplication gives

$$\frac{\prod_{v\in\{i,j\},\, w\in\{k,l\}}(1 - z_v z_w)}{Z_4} = R + t_i t_j + t_k t_l, \qquad R = Z_4 + Z_4^{-1} + 2 - \sum_{v<w} \big(z_v z_w + (z_v z_w)^{-1}\big).$$

Multiplying the proposed identity by $\prod_{i<j}(1 - z_i z_j)$, the common R term cancels because $a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23} = 0$. The remaining left side, divided by Z_4, is

$$\begin{aligned} a_{12}a_{34}(t_1t_2 + t_3t_4) - a_{13}a_{24}(t_1t_3 + t_2t_4) \\ + a_{14}a_{23}(t_1t_4 + t_2t_3) = \det[\,1,\ y_i,\ t_i,\ y_i t_i\,]_{i=1}^{4}. \end{aligned}$$

Here the last equality is a two-column Laplace expansion. Since $t_i = 2S - y_i - y_i^{-1}$, column operations give

$$\det[1, y_i, t_i, y_i t_i] = \det[1, y_i, y_i^{-1}, y_i^2] = \frac{\det[y_i, y_i^2, 1, y_i^3]}{\prod_i y_i} = \frac{\prod_{i<j}(y_i - y_j)}{\prod_i y_i}.$$

This is the required cleared numerator. □

Lemma 4.2 (Pfaffian propagation)**.** *Suppose a skew matrix K has the property that every four-index principal submatrix satisfies*

$$K_{ab}K_{cd} - K_{ac}K_{bd} + K_{ad}K_{bc} = \prod_{v<w\in\{a,b,c,d\}} K_{vw} \quad (a < b < c < d).$$

Then, at every even order for which the intervening products can be canceled, $\operatorname{Pf} K = \prod_{i<j} K_{ij}$. *For a rational or analytic family with a dense set of such points, the identity extends to the whole family.*

Proof. Let I be the first $N-4$ indices and let a, b, c, d be the last four. The elementary Pfaffian condensation identity is

$$\operatorname{Pf} K_{Iabcd} \operatorname{Pf} K_I = \operatorname{Pf} K_{Iab} \operatorname{Pf} K_{Icd} - \operatorname{Pf} K_{Iac} \operatorname{Pf} K_{Ibd} + \operatorname{Pf} K_{Iad} \operatorname{Pf} K_{Ibc}.$$

When $K_I = A$ is invertible, simultaneous row and column elimination in $\left[\begin{smallmatrix} A & B \\ -B^T & C \end{smallmatrix}\right]$ has determinant one and gives the diagonal blocks A and $C + B^T A^{-1} B$. Pfaffian multilinearity therefore reduces the displayed formula to the defining four-by-four Pfaffian formula. The identity is polynomial, so it holds also when A is singular.

Assume the product formula at all smaller even orders. Writing $P_I = \prod_{i<j\in I} K_{ij}$, the right side factors as

$$P_I^2 \prod_{i\in I,\, v\in\{a,b,c,d\}} K_{iv}(K_{ab}K_{cd} - K_{ac}K_{bd} + K_{ad}K_{bc}) = P_I \prod_{i<j} K_{ij}.$$

The left side is $P_I \operatorname{Pf} K$. Cancel P_I. The orders two and four start the induction. □

Proposition 4.3 (Complete product and local coefficient)**.** *Choose local square roots $r_i^2 = z_i/y_i$ and set $K_{ij} = -ir_i r_j B_{ij}$. For every even N on the dispersion curve,*

$$\operatorname{Pf} K = \prod_{i<j} K_{ij}, \qquad \det K = \prod_{i<j} K_{ij}^2.$$

The physical kernel has precisely this local form. The determinant identity is independent of all square-root choices.

Proof. At order four,

$$\operatorname{Pf} K = -\Big(\prod_i r_i\Big) \operatorname{Pf} B = -\Big(\prod_i r_i^3\Big) \prod_{i<j} B_{ij} = \prod_{i<j} K_{ij}.$$

The preceding identity verifies the four-point hypothesis for K. Apply Lemma 4.2 at generic points, where all pair entries are nonzero, then clear denominators and extend by analytic continuation to degenerate parameter values. The identity $(\operatorname{Pf} K)^2 = \det K$ gives the second equality. Changing a square root changes one row and its column by a minus sign and leaves both determinant expressions invariant. □

Corollary 4.4 (Fixed-order collision divisibility)**.** *On every regular fixed-order collision chart, each pair entry has a factor $\theta_i - \theta_j$; permutation covariance of the coupled contour gives the same factor when occupancies are present. Thus* $\det K$ *has the full squared Vandermonde factor, with bounded fixed-order jets on compact regular charts. Near $\theta_i = -\alpha + u_i$, $s = e^{i\alpha}$, the physical dispersion is $\gamma_i = i\alpha + O(u_i)$, and hence*

$$K_{ij} = \frac{u_i - u_j}{2i \sin\alpha} + O((u_i - u_j)|u|), \qquad \det K = \Delta(u)^2 \left[(2i\sin\alpha)^{-N(N-1)} + O(|u| + |s - e^{i\alpha}|)\right].$$

The analytic quotient and every parameter derivative retain the same Vandermonde factor. With all one-body measures, $1 + Z \to 2$, and the factor $1/N!$ included, the normalized local coefficient is exactly

$$A_\alpha = \frac{2}{N!(2\pi)^{N-1}} 2^{-N(N-1)} \sin(\alpha)^{-N^2} > 0.$$

Indeed the combined phase is $(-i)^N(-1)^{N(N-1)/2} = i^{-N^2} = 1$ for even N.

5 Deformation preserving the total angle

We use the normalized coefficients C_N of (2.3). Fix a selected point s_*, put $S_* = s_* + s_*^{-1} \in (0,2)$, and write

$$s_\varepsilon = (1+\varepsilon)s_*, \qquad b = \arccos(S_* - 1), \qquad \xi = e^{-ib}, \qquad H = \log(1/\varepsilon).$$

The constants in this section may depend on the selected point and a fixed derivative order k, but never on the even particle number N or on ε. The infinite-tail estimates concern $N \geq N_* + 2$, where $k = N_*^2/2 - 1$. Set $N_c = 2k + 4$. The logarithmic absolute bounds through order k are smaller than the first singular term. The deformation and transport formulas also apply to the fixed even orders $2 \leq N \leq N_*$ in Section 10.

For the regular dispersion coordinate we write

$$\varphi = \arccos W, \qquad z = e^{-i\varphi}, \qquad \gamma = i\varphi.$$

Its physical continuation lies in the fourth quadrant near either inverse-dispersion branch. The angular integration coordinates remain θ_i, with $\sum_i \theta_i = 0 \pmod{2\pi}$.

Choose a smooth periodic function p supported in $(0,\pi)$, with $0 \leq p \leq 1$ and $p = 1$ on $[\delta, \pi - \delta]$. Here $\delta > 0$ is smaller than $\alpha/2$, $b/2$, and $(2\alpha - b)/2$, and is small enough that b is inside this plateau and $1 + g'(\theta) > 0$ on the upper endpoint sliver. Here $g(\theta) = \arccos(S_* - \cos\theta)$; the choices are possible because $b < 2\alpha$ and $g'(0) = 0$. Put $m(\theta) = p(-\theta)$. Thus p, m have disjoint supports, and both inverse branches have fixed plateau neighborhoods. Define

$$\begin{aligned} P &= \sum_i p(\theta_i), \qquad Q = \sum_i m(\theta_i), \\ h_i &= \frac{\tau}{N}\{Pm(\theta_i) - Qp(\theta_i)\}, \qquad y_i(\lambda) = e^{i\theta_i + \lambda h_i}, \quad 0 \leq \lambda \leq 1. \end{aligned} \tag{5.1}$$

The positive number τ is fixed and sufficiently small. In particular $\sum_i h_i = 0$, so $\prod_i y_i = 1$ throughout the deformation.

Lemma 5.1 (Exact constrained deformation)**.** *The full integral defining C_N is unchanged by (5.1). Its angular Jacobian and its fixed jets have bounds $C_j^N N^{C_j}$, uniformly in N.*

Proof. For $y = e^{v+i\theta}$,

$$\Im W = \Im S - \sinh v \sin\theta.$$

On the upper semicircle the motion in (5.1) is inward, and on the lower semicircle it is outward. Both motions increase $\Im W$. For the admissible exterior parameters it remains positive, so every z_i stays on the interior sheet. Thus $1 - Z$ and all the pair denominators $1 - z_i z_l$ stay nonzero. The rational expression (2.2) has already canceled the factor $1 - y_i y_l$. The closed constrained torus therefore undergoes a holomorphic homotopy.

Write $w_i = \theta_i - ih_i$, so $\sum_i w_i = \sum_i \theta_i$. The full matrix $I - iDh$ preserves the tangent hyperplane and induces the identity on its one-dimensional quotient. Its full determinant is consequently its constrained tangent determinant. Directly,

$$Dh = \frac{\tau}{N}\operatorname{diag}(Pm_i' - Qp_i') + \frac{\tau}{N}(m \otimes p' - p \otimes m').$$

The diagonal and the two rank-one terms have operator norms at most $C\tau$, independently of N. Hence the Jacobian has modulus at most C^N, and its inverse matrix is uniformly bounded after reducing τ. Differentiating the determinant, or its logarithmic derivative formula, shows that fixed jets add only a polynomial in N. □

Let η be smooth, $0 \le \eta \le 1$, with $\eta(t) = 0$ for $t \le 1/4$ and $\eta(t) = 1$ for $t \ge 1/2$. Partition the final deformed integral with the three weights

$$w_F = \eta(P)\eta(Q), \qquad w_L = \tfrac{1}{2}(1-\eta(P))(1+\eta(Q)), \qquad w_U = \tfrac{1}{2}(1-\eta(Q))(1+\eta(P)).$$

Their sum is one. Reflection $\theta \mapsto -\theta$ exchanges P, Q, takes y_i to y_i^{-1}, preserves z_i, and changes K to $-K$. For even N its determinant and the normalized constrained integral are unchanged. Therefore

$$C_N = F_N + 2L_N, \tag{5.2}$$

where F_N and L_N are the final integrals with weights w_F and w_L. All weights are imposed after the full closed homotopy. The support of L_N has $P \le 1/2$; it excludes the upper inverse branch and consists of the lower semicircle together with two small upper endpoint slivers.

5.1 The sector containing both branch populations

Lemma 5.2. *For every fixed $j \ge 0$ and every even $N \ge 4$,*

$$|F_N^{(j)}(s_\varepsilon)| \le C_j N^{3j+1} \left(\frac{C_j}{\sqrt{N}}\right)^N.$$

In particular these derivatives have a uniformly bounded absolute sum.

Proof. On this support $P, Q \ge 1/4$. Both bump supports stay a fixed distance from the endpoints $0, \pi$. The imaginary-part formula, and the bounded range of W, imply

$$-\log|Z| \ge c\left(N\varepsilon + \frac{\tau PQ}{N}\right) \ge \frac{c}{N}.$$

Every true lower branch has outward displacement $\tau P/N$, and every true upper branch has inward displacement $\tau Q/N$. Their normal forms give a branch distance at least c/N in the W plane. All other coordinates stay in fixed regular neighborhoods. Consequently $|\partial_s \log z_i| \le C\sqrt{N}$ on a disk $|s - s_\varepsilon| \le c/N^3$. Reduce c once: the variation of $\log Z$ is at most $CN^{3/2}c/N^3$, smaller than half its preceding gap. The true branches remain on their interior sheets throughout this disk.

We estimate the correlation determinant directly. Entries with both points near the same inverse branch are bounded: the common factor $\varphi_i + \varphi_l$ cancels, leaving $(\varphi_i - \varphi_l)$ times a bounded analytic coefficient. Away from an opposite-branch pair, compactness gives the same bound. Let B be the number of coordinates in the two fixed true-branch neighborhoods. If a pair is near opposite branches, its numerator is bounded and

$$|1 - z_i z_l| \ge c\sqrt{(P+Q)/N}, \qquad |K_{il}| \le C\sqrt{N/(P+Q)} \le C\sqrt{N/B}.$$

Indeed both regular branch coordinates remain in a common sector of width strictly less than π (a small fixed enlargement of the fourth quadrant suffices), while their imaginary dispersion margins are at least $c\tau P/N$ and $c\tau Q/N$. Each such coordinate contributes one to $P + Q$. There are at most B exceptional entries in any row; all remaining entries are bounded. Every row therefore has Euclidean norm at most $C\sqrt{N}$. Hadamard gives

$$|\det K| \le C^N N^{N/2}$$

uniformly on the disk.

The one-body factors have a common independent angular majorant

$$f(\theta) = C\left(1 + \operatorname{dist}(\theta, \{b, -b\})^{-1/2}\right).$$

To verify this, use the imaginary part of the branch normal form when $|u| \le C(\varepsilon + |h|)$ and its linear real part outside that interval. The small parameter disk is absorbed in the positive branch displacement. Thus $f \in L^{4/3}(\mathbb{T}) \cap L^1(\mathbb{T})$. With normalized circle measure, the constrained product integral is $f^{*N}(0)$. Indeed, $g = f * f$ belongs to L^2 by Young's inequality, and $g * g$ is bounded and continuous by Cauchy–Schwarz and continuity of translations in L^2. Further convolution with $f \in L^1$ preserves these properties. Thus evaluation at zero is legitimate, and Young's inequalities give

$$f^{*N}(0) \le \|f\|_{4/3}^4 \|f\|_1^{N-4} \le C^N, \qquad N \ge 4.$$

The Jacobian costs C^N by Lemma 5.1. The remaining product kernel costs CN. Division by $N! \ge (N/e)^N$ bounds the integral on the disk by $N(C/\sqrt{N})^N$. Cauchy's inequality proves the assertion. □

6 A regular coordinate for the one-sided contour

The support of L_N can be lifted to one angular interval $(-\pi - \delta, \delta)$: the remaining upper plateau has zero weight because $p(\theta_i) = 1$ would imply $P > 1/2$. The possible half-angle sign changes are row and column gauge changes, so this lift preserves the determinant and its integral. We use $w_i = \theta_i - ih_i$ and the physical phase $\varphi_i = \arccos(S - \cos w_i)$ throughout.

Let $N_c = 2k + 4$. Choose a fixed branch width, decreased further below if needed, so that

$$\delta_B < \frac{1}{4} \min_{\substack{2 \le n \le N_c \\ n \text{ even}}} \operatorname{dist}(-b, 2\pi\mathbb{Z}/n). \tag{6.1}$$

The minimum is positive by Lemma 3.1. The right interval will be $[-b - \delta_B, \delta]$; its complement on the one-sided support has a compact left root. Choose the displacement strength after fixing these margins and cutoffs.

Let

$$w_B(s) = -\arccos(S(s) - 1), \qquad q = \varphi + w - w_B(s).$$

The branch of w_B is fixed by $w_B(s_*) = -b$.

Lemma 6.1 (Regular inverse and right-side convexity)**.** *The one-sided limiting dispersion curve has a fixed complex neighborhood on which q is a regular coordinate and $w = U(s, q)$ is analytic, with uniformly bounded fixed jets. The inverse can be chosen on a fixed convex neighborhood $\mathcal{D}_R$ of the right q-interval, enlarged to contain a sufficiently short negative branch segment, so that*

$$U(s, 0) = w_B(s), \qquad U_q(s, 0) = 0, \qquad \operatorname{Re} U_{qq}(s, q) \ge c > 0 \quad (q \in \mathcal{D}_R).$$

The right angular interval extends from $-b - \delta_B$ to a sufficiently small positive endpoint sliver. The smallness of δ_B and the deformation strength guarantees that its actual q-images lie in $\mathcal{D}_R$.

Proof. On the limiting compact right arc let $g(\theta) = \arccos(S_* - \cos\theta)$. Choose the upper endpoint small enough that $1 + g'(\theta) > c > 0$. Since

$$g''(\theta) = -\frac{S_*(1 - \cos\theta \cos g(\theta))}{\sin^3 g(\theta)} < 0,$$

the coordinate $q = g(\theta) + \theta + b$ is strictly increasing. Near the branch, use φ itself: the lower inverse angular function is even analytic and $w - w_B = O(\varphi^2)$, so $q = \varphi + O(\varphi^2)$ is a regular analytic coordinate.

There is also an explicit right inverse. Put $v = q + w_B(s)$; then $v = w + \varphi$ and

$$U(s,q) = \frac{v}{2} - \arccos\left(\frac{S(s)}{2\cos(v/2)}\right). \tag{6.2}$$

The branch is fixed by $w < \varphi$ on the limiting right arc. Its real v-interval is contained strictly in $(-2\alpha, 2\alpha)$, after the upper sliver is shortened. With $x = v/2$ and $D = 4\cos^2 x - S_*^2 > 0$, differentiation gives

$$U_{qq}(s_*, q) = \frac{S_*}{4\cos^2 x\sqrt{D}} + \frac{S_* \sin^2 x}{D^{3/2}} > 0. \tag{6.3}$$

The endpoint at the lower branch is regular and has value $1/\sin b$. Compactness supplies a positive lower bound on the whole closed real q-interval. A sufficiently thin fixed convex rectangle about that interval retains $\operatorname{Re} U_{qq} \geq c$, also for s in a small complex neighborhood of s_*. At $q = 0$, (6.2), or the dispersion identity, gives $U = w_B$ and $U_q = 0$ identically in s.

For the remaining compact left arc the limiting phase is $\varphi = -\mathrm{i}\gamma$, $\gamma > 0$. There q has real part $\theta + b$, so its image is injective, and $\mathrm{d}q/\mathrm{d}w = 1 + g'(w)$ cannot vanish: g' is purely imaginary. The two endpoint slivers remain regular after a small extension. The left image has negative real part, the right image positive real part, and they meet only in the regular branch chart. A finite cover by analytic inverse neighborhoods therefore gives one inverse along this embedded compact arc. Shrinking the neighborhoods, then fixing the small deformation and parameter neighborhood, gives the asserted bounded jets. The right inverse is the specific branch of (6.2). □

Set

$$\phi(s,q) = q - U(s,q) + w_B(s), \qquad a(s,q) = \frac{U_q(s,q)}{\sinh(\mathrm{i}\phi(s,q))}.$$

The coefficient a is analytic and bounded at $q = 0$: both its numerator and denominator have a simple zero there, and $U_q = q/\sin b + O(q^2)$ at $s = s_*$. It and its fixed jets are bounded on the complete one-sided inverse neighborhoods.

The complete kernel is analytic in these regular coordinates. Near the double branch the even inverse in φ cancels the factor $\varphi_i + \varphi_j$ in its denominator. Elsewhere that denominator is separated, since the only limiting pair-product zero on the one-sided support is the double lower branch. In particular on $\mathcal{D}_R^2$,

$$|K(q_i, q_j)| \leq C|q_i - q_j|, \tag{6.4}$$

and fixed jets are bounded. The diagonal zero holds in the single inverse coordinate q on this domain, including its angular endpoint.

6.1 Square-root geometry on the physical contour

Lemma 6.2 (Branch normal form)**.** *Put $u = \theta + b$ and $a_B = \sin b > 0$. On the original unit contour, and on a true lower branch displaced outward by a common $t_0 \geq 0$, set $E_0 = \varepsilon + t_0$. Uniformly for the fixed small branch arc and $0 \leq t_0 \leq \tau$, one has*

$$\begin{aligned} &D = 1 - W = a_B u - i\beta + R, \qquad && cE_0 \leq \beta \leq CE_0, \\ &R \in \mathbb{R}, \quad |R| \leq C(u^2 + E_0^2), \qquad && D_u = a_B + O(|u| + E_0). \end{aligned}$$

Here β may depend smoothly on u, with $|\beta_u| \leq CE_0$. On the physical fourth-quadrant branch,

$$\varphi = \sqrt{2D}\, h(D), \qquad h(0) = 1, \qquad |\varphi'| \asymp (|u| + E_0)^{-1/2}, \tag{6.5}$$

where h is analytic. Moreover

$$-\Im\varphi \ge c\sqrt{|u|+E_0} \quad (u\le 0), \qquad \Re\varphi' \ge \tfrac{2}{3}|\varphi'| \quad (u\ge 0), \qquad \int_{|u|\le\delta_B} |\varphi'(u)|\,du \le C.$$

For coordinates with the same outward shift, a minimum nonnegative coordinate u_j satisfies $|\varphi_i'| \le C|\varphi_j'|$ for every other nonnegative coordinate.

Proof. Use $y = e^{t_0+i(-b+u)}$ and the exact real and imaginary parts of $W = S - \cosh(t_0 + i(-b+u))$. The imaginary part is

$$\beta = \Im S + \sinh(t_0)\sin(b-u),$$

and $\Im S = 2\sin(\arg s_*)\varepsilon + O(\varepsilon^2)$. The real part has the asserted expansion because $1 - S_* + \cos b = 0$, $\Re S - S_* = O(\varepsilon^2)$, and $\cosh t_0 - 1 = O(t_0^2)$. The analytic factorization of $\arccos(1-D)$ gives (6.5). The relative error with respect to $a_B u - i\beta$ is uniformly small after the branch width and τ are fixed small. Its square root lies in a small fixed enlargement of the fourth quadrant. For $u \le 0$ its imaginary part has size at least a fixed multiple of its modulus; for $u \ge 0$ the derivative lies in a cone whose angle is at most $\pi/4 + o(1)$ from the positive real axis. This proves the stated attenuation and $2/3$ cone bounds. The integrable majorant and the minimum-coordinate comparison follow immediately from the two-sided derivative estimate. □

6.2 Coordinate size and angular diameter

Lemma 6.3. *On the actual right contour put $u_i = \theta_i + b$, $t = N^{-1}\sum_i u_i$ and $d = \max_i \theta_i - \min_i \theta_i > 0$. For any r with $r \le d \le 2r$, set $T = |t| + r + \varepsilon$. Then*

$$|1 + \varphi_w(w_i)| \le C(|u_i| + \varepsilon)^{-1/2}, \tag{6.6}$$

$$\int_{t-2r}^{t+2r} (|u| + \varepsilon)^{-1/2}\mathrm{d}u \le \frac{Cr}{\sqrt{T}}, \tag{6.7}$$

and

$$|q_i| \le C\sqrt{T}, \qquad |q_i - q_l| + |K_{il}| \le Cr/\sqrt{T}. \tag{6.8}$$

These are pointwise estimates. No extrema, diameter, or mean scale is differentiated, and no partition of the integral is made here.

Proof. Near the lower branch, $p = 0$, $m = 1$, so the outward displacement is the same $\tau P/N$ at every such coordinate. The branch normal form gives (6.6); away from that branch compactness gives the same bound, with a larger constant. It also holds along a one-variable angular curve with the actual occupancies held fixed. The curve derivative has the additional bounded factor $1 - \mathrm{i}\tau[(P/N)m' - (Q/N)p'] = 1 + O(\tau)$.

If $|t| \ge 4r$, the denominator in (6.7) is comparable to $\sqrt{|t| + r + \varepsilon}$. Otherwise enlarge the interval to $[-6r, 6r]$ and use

$$\sqrt{6r+\varepsilon} - \sqrt{\varepsilon} = \frac{6r}{\sqrt{6r+\varepsilon} + \sqrt{\varepsilon}}.$$

This proves the interval estimate even when the interval crosses the branch. Joining two actual angles along their fixed-occupancy curve gives exactly the actual q endpoints. Integrating its derivative, and then using (6.4), proves the second inequality in (6.8).

To prove the absolute coordinate bound, first suppose $P = 0$. Nonnegativity of p makes every $p(\theta_i) = 0$, hence every $h_i = 0$. The branch normal form and compactness give $|\varphi_i| \le C\sqrt{|u_i| + \varepsilon}$. Since $q_i = \varphi_i + u_i - b - w_B(s)$ and $w_B(s) + b = O(\varepsilon)$, this implies $|q_i| \le C\sqrt{|u_i| + \varepsilon} \le C\sqrt{T}$. If $P > 0$, some actual right angle lies in the positive support of p. Therefore $\max u_i \ge b$ and $|t| + 2r \ge b$, so $T \ge b/2$. Boundedness of the fixed regular domain gives the same conclusion. All estimates allow arbitrary actual occupancy means $P/N, Q/N$ in $[0, 1]$ on the right rectangle. □

7 Tangential differentiation of the complete one-sided integral

7.1 The scalar density and its full Jacobian

Let $M = D_\theta w = I - \mathrm{i}Dh$. Its inverse is uniformly bounded for small τ, $\det M$ has bound C^N, and fixed jets add polynomial factors in N. Also $\mathbf{1}^T M = \mathbf{1}^T$, hence $\mathbf{1}^T M^{-1} = \mathbf{1}^T$. For $\chi_s(\theta) = (q_i)$,

$$D_\theta \chi_s = \mathrm{diag}(1/U_{q,i})M, \quad J = \det D_\theta \chi_s = \det M \prod_i (1/U_{q,i}). \tag{7.1}$$

This identity is used at $\varepsilon > 0$, where all factors are nonzero. The apparent branch poles will be retained as integrable Jacobian factors, never differentiated as unregularized amplitudes.

The complete deformed scalar density is $Jf(s,q)$, where

$$f(s,q) = \frac{1}{N!(2\pi)^{N-1}} \frac{1+Z}{1-Z} \det K \prod_i a(s,q_i). \tag{7.2}$$

Formula (7.2) includes the unchanged factor 2π converting the ambient normalized density to constrained Haar normalization. The entire occupancy Jacobian has cancelled from the regular amplitude.

For a residual field B in q coordinates define

$$V = -M^{-1}(\dot{U} + U_q B), \tag{7.3}$$

where a dot denotes the fixed-q derivative ∂_s and products inside the vector are componentwise. Differentiating $U(s, \chi_s(\theta)) = w(\theta)$ shows that (7.3) is exactly $(D_\theta \chi_s)^{-1}(\partial_s \chi_s - B)$. If

$$\sum_i U_{q,i} B_i = -\sum_i \dot{U}_i, \tag{7.4}$$

then $\sum_i V_i = 0$ on an ambient neighborhood. If in addition

$$\sum_i B_i = -N w_B'(s), \tag{7.5}$$

then $(\partial_s - V \cdot \nabla_\theta) \sum_i \varphi_i = 0$. Indeed $\sum \varphi_i = \sum q_i - \sum w_i + N w_B(s)$, and $\sum w_i = \sum \theta_i$ is frozen by tangency.

For every fixed smooth real angular weight ω the exact identity is

$$\partial_s(\omega J f) - \mathrm{div}_\theta(V \omega J f) = \omega J[\partial_s f + \mathrm{div}_q(Bf)] - (V \cdot \nabla_\theta \omega) J f. \tag{7.6}$$

To verify the identity, put $A_\chi = D_\theta \chi_s$. Commuting mixed derivatives of χ_s gives

$$(\mathrm{div}_q B) \circ \chi_s = \mathrm{tr}\left[A_\chi^{-1} \left(\partial_s A_\chi - \sum_i V_i \partial_i A_\chi \right) \right] - \mathrm{div}_\theta V.$$

Jacobi's formula therefore gives $J(\mathrm{div}_q B) \circ \chi_s = \partial_s J - \mathrm{div}_\theta(JV)$. The product and chain rules prove (7.6). Because $\sum V_i = 0$ identically, the intrinsic divergence on $\sum \theta_i = c$, parametrized by the first $N-1$ angles, is

$$\sum_{i<N} (\partial_i - \partial_N) V_i = \sum_i \partial_i V_i.$$

The same formula holds for V multiplied by any scalar density. Thus ambient divergence restricts to constrained divergence without an additional measure factor. Integration by parts applies to the real and imaginary parts of these complex fields. The identity acts on scalar coefficients relative to the original real constrained measure on $\mathbb{H}_N$. The cutoffs and

occupancies are smooth functions of the real angles. The commuting frame $(D_\theta \chi_s)^{-1}\nabla_\theta$ represents q differentiation. With $A_\chi = D_\theta \chi_s$, its operators are

$$E_a = \sum_i (A_\chi^{-1})_{ia}\partial_{\theta_i}, \qquad E_s = \partial_s - \sum_i (A_\chi^{-1}\partial_s \chi_s)_i \partial_{\theta_i}.$$

Then $E_a \chi_b = \delta_{ab}$ and $E_s \chi_b = 0$. Every commutator $[E_a, E_b]$ or $[E_s, E_a]$ is a linear combination of the real angular derivatives and annihilates all χ_b. Invertibility of A_χ therefore makes every commutator zero on the real angular domain.

7.2 A bounded transport when a left root is present

Choose a single smooth right cutoff $r(\theta)$, equal to one for $\theta + b \geq -\delta_B/2$ and zero for $\theta + b \leq -\delta_B$, with its harmless periodic return inside the excluded upper plateau. Decompose the complete one-sided integral by

$$1 = \prod_i r(\theta_i) + \left(1 - \prod_i r(\theta_i)\right).$$

Telescope the second term. Each of its at most N terms has one fixed named anchor a in a compact left region, including on every cutoff jet. That anchor has $|z_a| \leq q_0 < 1$, and $|U_{q,a}| \geq c > 0$.

On such a term take

$$B_a = -\frac{\sum_i \dot{U}_i}{U_{q,a}}, \qquad B_i = 0 \ (i \neq a). \tag{7.7}$$

Equation (7.4) holds; equation (7.5) is unnecessary because $|1 - Z| \geq 1 - q_0$. All fixed regular-coordinate and parameter jets of B, V and the regular amplitude have polynomial N bounds, apart from fixed one-body products costing C_j^N. In particular the derivatives of $Z = \exp(-\mathrm{i}\sum \phi_i)$ are bounded by a polynomial in N, since $|Z| \leq 1$ at the actual contour and fixed jets of each ϕ_i are bounded.

Hadamard applied to the complementary minors after fixed derivatives therefore bounds the transported density by

$$C_j^N N^{-N/2+C_j} \prod_i |1 + \varphi_w(w_i)|. \tag{7.8}$$

The named compact factor is bounded. Eliminate its angle using the constraint; every remaining factor has the independent majorant $C(1 + \mathrm{dist}(\theta, -b)^{-1/2})$. Integrating the product costs C^N. This proves a uniform summable bound for every fixed derivative of the whole left-root part, including all its mixed branch configurations.

7.3 One collective field on the entire right part

For this part all inverse coordinates lie in $\mathcal{D}_R$. Put

$$c_0 = -w_B'(s), \quad A = -\sum_i \dot{U}_i - c_0 \sum_i U_{q,i}, \quad R = \sum_{i<j} (U_i - U_j)^2,$$

and, off full equality,

$$B_i = c_0 + \frac{A}{R} \sum_{j \neq i} \frac{(U_i - U_j)^2}{U_{q,i} - U_{q,j}}. \tag{7.9}$$

Each summand extends analytically through its pair diagonal. Indeed

$$E(s; q, r) = \frac{U_q(s, q) - U_q(s, r)}{q - r} = \int_0^1 U_{qq}(s, r + t(q - r))\, \mathrm{d}t$$

has $\operatorname{Re} E \geq c$; hence, writing $H = (U(s,q) - U(s,r))/(q-r)$, the summand is $(q-r)H^2/E$. Pairing its antisymmetric terms gives

$$\sum_i B_i = Nc_0 = -Nw_B', \qquad \sum_i U_{q,i} B_i = c_0 \sum_i U_{q,i} + A = -\sum_i \dot{U}_i. \tag{7.10}$$

Thus one field freezes both required quantities on the entire right part, including every mixture of branch and compact coordinates.

Lemma 7.1 (Uniform fixed-coordinate jets)**.** *Let $d = \max \theta_i - \min \theta_i$. On the actual right contour and for any fixed total jet order,*

$$|\partial_s^a \partial_q^\beta B_i| + |\partial_s^a \partial_q^\beta V_i| \leq C_j N^{C_j} d^{-1-|\beta|}. \tag{7.11}$$

There is no additional inverse power of ε.

Proof. First, at actual contour points, $U_i - U_j = w_i - w_j = (\theta_i - \theta_j) - \mathrm{i}(h_i - h_j)$. The same occupancies occur in both h_i, h_j, so

$$|h_i - h_j| \leq C\tau |\theta_i - \theta_j|, \quad \operatorname{Re} R \geq (1 - C^2\tau^2) \sum_{i<j} (\theta_i - \theta_j)^2. \tag{7.12}$$

Consequently $|R| \geq cd^2$ and $\sum_{i<j} |U_i - U_j|^2 \leq C|R|$.

To estimate parameter derivatives in terms of the diameter, use $U(s,0) = w_B(s)$ and $U_q(s,0) = 0$ on the fixed convex domain to obtain

$$|\partial_s^a U_q(s,q)| \leq C_a |q|. \tag{7.13}$$

Apply Lemma 6.3 with $r = d$ and write $T_d = |t| + d + \varepsilon$. Every point of the straight segment joining q_i and q_j has modulus at most $C\sqrt{T_d}$. Integration of (7.13) along this segment gives

$$|\partial_s^a (U_i - U_j)| \leq C_a \sqrt{T_d}\, |q_i - q_j| \leq C_a d. \tag{7.14}$$

Consequently $|\partial_s^a R| \leq C_a N^2 d^2$. Its first fixed-$q$ derivatives are $O(C_j N^{C_j} d)$, since each differentiated product retains a pair difference. Higher fixed coordinate derivatives are bounded by $C_j N^{C_j}$. These assertions include arbitrary fixed additional parameter derivatives, by (7.14) and bounded analytic jets. Together with $|R| \geq cd^2$, the quotient rule yields

$$|\partial_s^a \partial_q^\beta R^{-1}| \leq C_j N^{C_j} d^{-2-|\beta|}.$$

For $d \leq 1$, this follows by assigning a factor $d^{2-|\beta|}$ to each jet of R; fixed parameter derivatives cost no power. For $d \geq 1$, the denominator is uniformly separated and the fixed maximum diameter is absorbed in the constant.

Every fixed parameter jet of H has size $O(\sqrt{T_d})$ by (7.13), whereas all fixed jets of E^{-1} are bounded. Thus the pair numerator $F(q_i, q_j) = (q_i - q_j)H^2/E$ satisfies

$$|\partial_s^a F(q_i, q_j)| \leq C_a |q_i - q_j| T_d \leq C_a d \sqrt{T_d} \leq C_a d.$$

Every positive coordinate-order jet is bounded, including additional fixed parameter derivatives, because F is analytic on the fixed product domain. It follows that

$$|\partial_s^a \partial_q^\beta \sum_{l \neq i} F(q_i, q_l)| \leq C_j N^{C_j} d^{1-|\beta|}.$$

All jets of A are polynomially bounded in N. Formula (7.9) now proves the stated bound for B.

For the angular representative, the exact frame coefficients are

$$E_a\theta_i = (M^{-1})_{ia}U_{q,a}, \qquad E_s\theta_i = (M^{-1}\dot{U})_i.$$

Their fixed jets are bounded by induction: differentiate M^{-1} by $D(M^{-1}) = -M^{-1}(DM)M^{-1}$, and use the bounded real angular derivatives of M and the bounded analytic jets of $U_q, \dot{U}$. At fixed order the index sums contribute a polynomial in N. The same induction bounds the jets of the real cutoffs. The frame formulas involve U_q with a nonnegative power. Applying the product rule to (7.3) gives the same bound for V and permits iteration of (7.6) with these cutoffs. □

At fixed ε, N, the complete product identity gives collision degree $N(N-1)$ in the original angular differences. The field has one inverse diameter and each transport loses at most two degrees. Thus the full-equality puncture flux vanishes through order $j \le k$ whenever

$$N^2 - 2j - 3 > 0, \qquad N \ge N_* + 2, \quad k = N_*^2/2 - 1. \tag{7.15}$$

The pair quotients in (7.9) are regular, so fluxes need be checked only at full equality. Apply the transports to the smooth right term before selecting extrema or diameter shells.

Differentiating a determinant marks a matrix entry and leaves a complementary minor to which Hadamard's inequality applies. At fixed order the number of index choices grows polynomially in N. Each field coefficient costs one inverse diameter, its coordinate derivative costs one more, and its parameter derivative preserves the power. The product rule and divergence give the following filtration by induction. a descendant after j transports contains ℓ marked entries and a scalar loss d^{-h} with

$$h + \ell \le 2j. \tag{7.16}$$

All regular entry, one-body and scalar jets are bounded with polynomial N costs. If the unchanged entries are at most a, Hadamard and $N! \ge (N/e)^N$ leave

$$C_j^N N^{-N/2+C_j} d^{-h} a^{N-\ell} \frac{\prod_i |1 + \varphi_w(w_i)|}{|1 - Z|}. \tag{7.17}$$

8 One coarea estimate for the complete right integral

8.1 A projected periodic kernel

Lemma 8.1 (Oblique periodic kernel)**.** *Fix $|\vartheta| < \pi/2$ and put $P_\vartheta = 2\pi\cos\vartheta$. If $G = x - ia$, $a \ge a_0$, and $0 < a_0 \le 1$, then, with $v = \Re(e^{-i\vartheta}G)$,*

$$\frac{1}{|1 - e^{-iG}|} \le \frac{C_\vartheta}{a_0 + \operatorname{dist}(v, P_\vartheta\mathbb{Z})}.$$

Consequently, on any injective v-parametrized arc whose v-image has length L, the integral of this majorant in v is at most $C_\vartheta(1 + L)(1 + |\log a_0|)$.

Proof. For $0 < a \le 1$, use

$$|1 - e^{-a-ix}|^2 = (1 - e^{-a})^2 + 2e^{-a}(1 - \cos x) \ge c\{a^2 + \operatorname{dist}(x, 2\pi\mathbb{Z})^2\}.$$

It follows that $|1 - e^{-iG}| \ge c\{a + \operatorname{dist}(x, 2\pi\mathbb{Z})\}$. For an integer n minimizing the latter distance,

$$|v - 2\pi n\cos\vartheta| \le \cos\vartheta\,|x - 2\pi n| + |\sin\vartheta|a.$$

Thus $a_0 + \operatorname{dist}(v, P_\vartheta\mathbb{Z})$ is bounded by a constant times the preceding lower bound. For $a \ge 1$, $|1 - e^{-iG}| \ge 1 - e^{-1}$ and the proposed denominator is bounded above by $1 + P_\vartheta/2$. Integrating over translates of one period gives the integral bound. □

For the actual contours the attenuation floor is $-\log|Z| \geq cN\varepsilon$. Indeed $\Im W_i \geq c\varepsilon$, and $W_i = \cos\varphi_i$ remains in a bounded set. Writing $\varphi_i = x_i - ia_i$ gives $\Im W_i = \sin x_i \sinh a_i \leq Ca_i$, hence $a_i \geq c\varepsilon$. Thus Lemma 8.1 costs at most $C(1+L)(1+|\log\varepsilon|)$ on an injective projected phase interval of length L.

8.2 Oblique extreme coarea with the full occupancy dependence

Let $I = [-b-\delta_B, \delta_R]$, where $\delta_R = \delta < b/2$, and choose the negative segment inside the lower branch plateau. The upper inverse branch is outside I. Use the original balanced contour with reflected disjoint bumps and put

$$\sigma = P/N, \quad \upsilon = Q/N, \quad f(u;\sigma,\upsilon) = -b + u - i\tau\{\sigma m(-b+u) - \upsilon p(-b+u)\}.$$

Here $u \in [-\delta_B, b+\delta_R]$ and the original one-sided occupancies satisfy $P \leq 1/2$, $0 \leq Q \leq N$. The auxiliary analytic estimates in terms of $E = \varepsilon + \tau\sigma$ hold on the fixed box $0 \leq \sigma, \upsilon \leq 1$. The final bound in Lemma 8.2 concerns actual occupancies computed from the angles, including on the enlarged rectangle used below. Let $\varphi_s(w) = \arccos(S(s) - \cos w)$ on the physical sheet, $b_s(w) = \partial_w \varphi_s(w)$. Use the regular coordinate defined above. The regularized phase coordinate is $q_s(w) = \varphi_s(w) + w - w_B(s)$. Its analytic inverse and the scalar transport were proved in Sections 6–7.

Lemma 8.2 (Occupancy-dependent extreme derivative)**.** *Write $G(\theta) = \sum_i \varphi_s(w_i(\theta))$ and $\vartheta = 3\pi/8$. On an actual extreme-index cell with $0 \leq m = \min_i u_i < M = \max_i u_i$, all other u_i fixed, and $m + M$ fixed by the constraint, one has*

$$\Re\left(e^{-i\vartheta}\frac{dG}{dm}\right) > 0, \qquad \frac{|(b_s(w_m)+1)(b_s(w_M)+1)|}{\Re(e^{-i\vartheta}dG/dm)} \leq C\frac{\sqrt{M+\varepsilon}}{M-m}. \tag{8.1}$$

The constants are independent of N, ε, and the actual occupancies. A sufficiently small fixed τ is chosen after the bump derivatives and fixed angular margins.

Proof. Temporarily hold the actual values of σ, υ fixed and set $E = \varepsilon + \tau\sigma$. On the plateau $f = -b + u - i\tau\sigma$; the square-root normal form gives

$$|b_s(f(u))| \leq C(u+E)^{-1/2}, \qquad |\varphi_s''(f(u))| \asymp (u+E)^{-3/2}, \qquad u \geq 0. \tag{8.2}$$

These estimates extend to the remainder of the fixed right interval by compactness: it stays a fixed distance from both dispersion branches, while u is bounded above and below there. The exact identity

$$-\varphi_s''(w) = \frac{S(s)(1 - S(s)\cos w + \cos^2 w)}{\sin^3\varphi_s(w)}$$

also gives

$$\Re\{e^{-i\vartheta}[-\varphi_s''(f(u))]\} \geq c|\varphi_s''(f(u))|. \tag{8.3}$$

Indeed on the true branch the normal form puts $\arg(\sin\varphi)$ in an arbitrarily small enlargement of $[-\pi/4, 0]$. The real branch-center displacement is $O(\varepsilon^2 + (\tau\sigma)^2)$ and the imaginary margin is comparable to E, so this statement includes $u = 0$. On the compact remainder $\sin\varphi$ is close to a positive real number. The numerator is uniformly close to a positive real number since $1 - S_*c + c^2 \geq 1 - S_*^2/4 > 0$ for real c. Thus (8.3) holds after fixing small τ, ε_0.

Put $a(u) = \partial_u f(u)$ and $B(u) = b_s(f(u))a(u)$ at these fixed occupancies. Since $a = 1 + O(\tau)$, $a' = O(\tau)$, and $|b_s(f)| \leq C|\varphi_s''(f)|$ by (8.2),

$$-B' = -\varphi_s''(f)a^2 - b_s(f)a'$$

retains (8.3). Therefore, with $d = M - m$,

$$\begin{aligned} &\Re\{e^{-i\vartheta}[B(m) - B(M)]\} \geq cL, \\ &L = \int_m^M (u+E)^{-3/2}du = \frac{2d}{\sqrt{m+E}\sqrt{M+E}(\sqrt{m+E}+\sqrt{M+E})}. \end{aligned} \tag{8.4}$$

Differentiate on the constraint with both occupancies varying, and set

$$B_m^{\mathrm{occ}} = \sum_v m(\theta_v)b_s(w_v), \qquad B_p^{\mathrm{occ}} = \sum_v p(\theta_v)b_s(w_v).$$

Direct differentiation of the original balanced contour gives the exact formula

$$\partial_{\theta_i} G = b_s(w_i)a(u_i) - \frac{i\tau p'(\theta_i)}{N}B_m^{\mathrm{occ}} + \frac{i\tau m'(\theta_i)}{N}B_p^{\mathrm{occ}}. \tag{8.5}$$

The constraint slice differentiates in the direction $\partial_{\theta_{\min}} - \partial_{\theta_{\max}}$. Every actual $u_v \geq m$, and the true branches have the same outward displacement $\tau\sigma$. Hence (8.2) implies

$$\frac{|B_m^{\mathrm{occ}}| + |B_p^{\mathrm{occ}}|}{N} \leq C(m+E)^{-1/2}.$$

Both bump-derivative differences in (8.5) are bounded by Cd. The total occupancy correction to $B(m) - B(M)$ is thus at most $C\tau d(m+E)^{-1/2}$. Since $M + E$ lies in a fixed bounded interval, the exact expression in (8.4) gives $d(m+E)^{-1/2} \leq CL$. Choosing τ small absorbs the correction into half the positive projection in (8.4). This compares the derivative at the actual extreme configuration. The auxiliary curve was used to compare the two frozen-occupancy endpoint values.

The numerator in (8.1) is at most $C[(m+E)(M+E)]^{-1/2}$, with the constant enlarged on the bounded interval. Combining with (8.4) bounds the ratio by $C\sqrt{M+E}/d$. If $P = 0$, then $E = \varepsilon$. If $P > 0$, an actual angle belongs to the positive support of p, so $M \geq b$. Therefore $E \leq C(M + \varepsilon)$ in either event, proving (8.1). □

The geometric integration domain. After all differentiation, bound the smooth weight and its transported coefficients by their proved envelopes and enlarge the nonnegative integral to the full right angular rectangle $I^N \cap \mathbb{H}_N$. All actual occupancies there still satisfy $0 \leq \sigma, \upsilon \leq 1$, so the preceding lemma applies. The balanced displacement has the same sign on each semicircle, and hence $\Im(S - \cos w_i) \geq \Im S$: the physical sheet and attenuation floor are preserved on this enlarged domain. The regular inverse also remains valid there. On the compact right arc its q-image changes by $O(\varepsilon + \tau)$; near the branch, uniform square-root continuity bounds the change by $O(\sqrt{\varepsilon + \tau})$. Choose δ_B, τ and ε_0 after the fixed margin of $\mathcal{D}_R$ so that all these images lie inside it.

For fixed other coordinates and a fixed constrained sum, the conditions specifying the extreme indices, a diameter shell, and a nonnegative minimum are linear interval conditions in the remaining variable m, with $M = A - m$. Their intersection is one interval. This is the enlarged geometric fiber after bounding the smooth cutoff. The estimate for a negative minimum is given below.

8.3 A fixed mean scale for every diameter

Write $u_i = \theta_i + b$, $t = N^{-1}\sum_i u_i$, and choose actual minimum m and maximum M only in the final absolute integral. The angular constraint fixes $t = b + 2\pi l/N$; at most CN such means meet the right interval. The independent coordinate measure after eliminating any one angle is unchanged.

On a shell $r \le d \le 2r$ use the scale $T = |t| + r + \varepsilon$ from Lemma 6.3. Every coordinate belongs to $[t - 2r, t + 2r]$.

For a nonnegative minimum, Lemma 8.2 gives the monotone projected phase and

$$\frac{|1 + \varphi_w(w_m)|\,|1 + \varphi_w(w_M)|}{|dv/dm|} \le C\frac{\sqrt{M + \varepsilon}}{M - m}. \tag{8.6}$$

Its phase range is at most CN, since each phase is bounded. Lemma 8.1 therefore costs $CN(H + 1)$. Here and below $H = |\log \varepsilon|$.

For a nonnegative minimum, (8.6) therefore bounds the integral in the remaining extreme coordinate by

$$\int \frac{|1 + \varphi_w(w_m)|\,|1 + \varphi_w(w_M)|}{|1 - Z|}\mathrm{d}m \le CN\frac{\sqrt{T}}{r}(H + 1). \tag{8.7}$$

For a negative minimum, m lies in the fixed true-branch segment, where its attenuation gives

$$\frac{|1 + \varphi_w(w_m)|\,|1 + \varphi_w(w_M)|}{|1 - Z|} \le \frac{C}{(|m| + \varepsilon)\sqrt{|M| + \varepsilon}}.$$

Set $a = |m| + \varepsilon$, $b_1 = |M| + \varepsilon$. Since $a + b_1 \ge M - m \ge r$, the elementary inequality

$$\frac{1}{a\sqrt{b_1}} \le \frac{1}{\sqrt{r}}\left(\frac{1}{a} + \frac{1}{b_1}\right) \tag{8.8}$$

follows by multiplying through by $a\sqrt{b_1}$ and using $a + b_1 \ge r$. On the constrained slice $M = A - m$, so each reciprocal in (8.8) integrates to $C(H + 1)$ over a bounded interval. This gives

$$\int \frac{|1 + \varphi_w(w_m)|\,|1 + \varphi_w(w_M)|}{|1 - Z|}\mathrm{d}m \le Cr^{-1/2}(H + 1) \le C\frac{\sqrt{T}}{r}(H + 1), \tag{8.9}$$

which is the same pair envelope as (8.7).

The two actual extreme Jacobian factors have been retained through (8.7); only the $N - 2$ remaining factors are now replaced by independent one-body majorants. Their interval integrations cost $(Cr/\sqrt{T})^{N-2}$. Combine this with (7.16), (7.17), (6.8), and (8.7)–(8.9): the complete shell contribution is bounded by

$$C_j^N N^{-N/2+C_j} r^{2N-2j-3} T^{-N+3/2}(H + 1) \le C_j^N N^{-N/2+C_j} r^{N-2j-3/2}(H + 1). \tag{8.10}$$

Here the fixed number of marked entries permits all bounded powers of T or r to be absorbed into C_j. The last inequality uses $T \ge r$. This estimate is valid for every $r > 0$.

For all sufficiently large N, specifically $N > 2k + 4$, the exponent in (8.10) is positive for every $j \le k$. Summing dyadic shells and the polynomial number of extrema and means gives a summable $C_j^N N^{-N/2+C_j}(H + 1)$ envelope.

For the finite orders $N_* + 2 \le N \le 2k + 4$, choose δ_B after the full finite list $2 \le n \le 2k + 4$, n even, so that the distance of every constrained mean from the branch exceeds $4\delta_B$. The same choice will be used in the finite-order complementary lemma. On the right interval $u \ge -\delta_B$, a sufficiently small-diameter configuration must then have every u_i in a fixed compact right interval away from zero. There the full fixed-N Vandermonde degree gives the shell power $N^2 - 2j - 3 > 0$. The remaining shells have a fixed positive lower diameter and (8.10) suffices, even if its exponent is negative. Thus the finite block has a uniform logarithmic bound as well.

9 The complete logarithmic tail

Theorem 9.1. *At each selected point and for every $0 \le j \le k$, uniformly for $0 < \varepsilon < \varepsilon_0$,*

$$\sum_{\substack{N>N_* \\ N \text{ even}}} |C_N^{(j)}((1+\varepsilon)s_*)| \le C_j(1+|\log\varepsilon|). \tag{9.1}$$

In particular the entire higher-particle contribution is $o(\varepsilon^{-1/2})$ at the derivative order of the first singularity.

Proof. The selected part has the summable bound proved in Lemma 5.2. The left-root part has the bound (7.8). For $N > N_c$, summing (8.10) over dyadic shells gives the same factorial envelope times $H+1$. For the finitely many orders $N_*+2 \le N \le N_c$, the finite-mean argument following (8.10) gives a finite constant times $H+1$. These estimates include every part of the exact identity $C_N = F_N + 2L_N$. Summing proves the theorem. Since $\varepsilon^{1/2}(1+|\log\varepsilon|) \to 0$, the tail is smaller than the first singular term.

The choices have the following order: the selected point, k and N_c; the fixed endpoint slivers and inverse-coordinate neighborhood; the finite mean gap and a sufficiently small branch width; the smooth bumps and right cutoff; a sufficiently small fixed τ; the protected parameter-disk constants; and finally ε_0. Each estimate holds uniformly in all tail orders after those choices. Divide into diameter ranges after transporting the complete weighted integrals. □

10 The first singularity on the constrained angular torus

Fix $\zeta = e^{i\alpha}$ from Lemma 3.1, and put

$$N = N_* = 2p, \qquad k = N^2/2 - 1, \qquad s_\varepsilon = (1+\varepsilon)\zeta, \qquad H = \log(1/\varepsilon).$$

Use the coefficients C_n in (2.3). The same deformation and reflection decomposition (5.2) apply for $2 \le n \le N$, since the deformation and field formulas hold at these orders.

The geometric choices already made give $\delta < \alpha/2$ and (6.1). Hence $-\alpha$ lies on $m = 1, p = 0$, $+\alpha$ lies on $p = 1, m = 0$, and no constrained full-equality configuration of the present finite orders lies in the chosen branch neighborhood.

10.1 Finite-order complementary regions

Lemma 10.1. *Every positive even $n < N_*$ has bounded derivatives of every fixed order along s_ε. At $N = N_*$, choose a fixed smooth shape cutoff equal to one near $\theta_1 = \cdots = \theta_N = -\alpha$, supported where the deformation is the identity and $w_L = 1$. Let I_N denote that part of L_N. Then $C_N - 2I_N$ has bounded derivatives of every fixed order along this approach.*

Proof. We first prove $O(1+H)$ for each fixed derivative. All constants here may depend on the fixed order n and the number of derivatives.

For F_n, both occupancies are at least $1/4$. Each true branch has displacement at least $\tau/(4n)$ and the product has a positive damping gap at s_*. All one-body and pair denominators are separated there: the only limiting possible unit pair-product zero on these physical contours is at a dispersion branch, and the displacements separate those branches. Compactness supplies a fixed parameter disk on which the localized integrand is bounded and holomorphic. This fixed-order compactness argument also applies at $n = 2$.

The left-root pieces of L_n have bounded fixed derivatives by the anchor construction of Section 7.2. Its argument is valid for each fixed $n \ge 2$ and every fixed number of derivatives: eliminate the named compact anchor, retain its bounded Jacobian, and integrate the other independent branch majorants. Thus only the complete right part remains.

At full equality the constraint requires $\theta_i = t_\theta = 2\pi\ell/n$. By (6.1), none of these finitely many points is at the branch. In the right interval their shifted means satisfy $t_\theta + b > 4\delta_B$, so the equality neighborhoods are compact regular right charts. At equality the balanced deformation is exactly zero, since $P = np(t_\theta)$ and $Q = nm(t_\theta)$ give $h_i = 0$. Consequently its product denominator can vanish only if

$$n \arccos(S_* - \cos t_\theta) \in 2\pi\mathbb{Z}, \qquad S_* = \cos(2\pi\ell/n) + \cos(2\pi m/n).$$

Lemma 3.1 excludes this for $n < N_*$. At $n = N_*$ it gives only $t_\theta = \pm\alpha$ modulo 2π; the positive point is outside the one-sided support. Thus its only critical equality is $-\alpha$.

Localize each noncritical equality in a fixed small neighborhood where the roots and pair factors are regular and $1 - Z$ is separated. Such localized integrals extend holomorphically through s_*. At N_* remove also the specified chart at $-\alpha$. It lies entirely in $m = 1, p = 0$, so its deformation is zero, $w_L = 1$, and its right cutoff is one. On the remaining right support, including every cutoff derivative, the diameter satisfies $d \geq d_0 > 0$.

The single field of Section 7.3 is now smooth with bounded fixed regular jets, for any chosen number of derivatives. There is no puncture flux. The scalar transport gives the pointwise bound

$$C_{n,j} \frac{\prod_i |1 + \varphi_w(w_i)|}{|1 - Z|}.$$

Retain the two extreme-coordinate factors for Lemma 8.2, then bound the remaining factors by independent majorants. For a negative minimum use (8.8) instead. There are finitely many means, extreme-index choices and diameter shells with $d \geq d_0$. Equations (6.7), (8.7) and (8.9) therefore give $O(1 + H)$ for every fixed derivative of this entire remainder, including $n = 2$ when there are no other coordinates. The enlargement to a geometric interval in Section 8 handles disconnected smooth-cutoff fibers.

Each complete remainder is holomorphic on the exterior sector: all contours and angular localizations are fixed independently of s. For such a remainder R and $0 < \varepsilon \leq \varepsilon_1$,

$$R^{(j)}((1+\varepsilon)s_*) = R^{(j)}((1+\varepsilon_1)s_*) - s_* \int_\varepsilon^{\varepsilon_1} R^{(j+1)}((1+t)s_*)\, dt.$$

The already proved bound at order $j + 1$ is integrable at zero, proving boundedness. Finally $C_{N_*} - 2I_{N_*} = F_{N_*} + 2(L_{N_*} - I_{N_*})$ fixes the reflection factor exactly. □

10.2 The local shape integral

Near the lower configuration write

$$\theta_i = -\alpha + t_i, \qquad t_N = -\sum_{i<N} t_i.$$

The angular constraint is exactly $\sum_i t_i = 0$, since $N\alpha \in 2\pi\mathbb{Z}$. Its normalized measure is $\prod_{i<N} dt_i/(2\pi)$. All angles here are negative and lie on $m = 1, p = 0$, so the deformation is identically zero and $w_L = 1$. The physical regular dispersion coordinate has the expansion

$$\varphi_i = \alpha - 2i\varepsilon + t_i - \cot\alpha\, t_i^2 + O(\varepsilon^2 + \varepsilon|t_i| + |t_i|^3).$$

Put $q(t, s) = 1 - Z(t, s)$. Then

$$\begin{aligned} q(t, s_\varepsilon) &= 2N\varepsilon - i\cot\alpha \sum_i t_i^2 + O(\varepsilon^2 + \varepsilon|t| + |t|^3), \\ \partial_s q(0, \zeta) &= 2N\zeta^{-1} =: D_s. \end{aligned} \tag{10.1}$$

The particle number is fixed in these expansions.

Lemma 10.2. *On a sufficiently small fixed real shape ball,*

$$|q(t, s_\varepsilon)| \geq c(\varepsilon + |t|^2).$$

Proof. Writing $\rho^2 = \sum_i t_i^2$, the leading part in (10.1) satisfies

$$|2N\varepsilon - i\cot\alpha\,\rho^2| \geq c_0(\varepsilon + \rho^2).$$

On $0 < \varepsilon < \varepsilon_0$ and $\rho < \delta_0$, its remainder is at most $C(\varepsilon_0 + \delta_0)(\varepsilon + \rho^2)$. Choose the two fixed radii so that this is at most half the preceding lower bound, and apply the triangle inequality. □

The local integral can be written exactly as

$$I_N(s) = \int_{\mathbb{R}^{N-1}} \frac{\chi(t)\Delta(t)^2 A(t,s)}{q(t,s)}\, dt_1 \cdots dt_{N-1}, \tag{10.2}$$

where χ is the fixed shape cutoff and A is analytic near $(0, \zeta)$. Its value is

$$A(0,\zeta) = A_\alpha = \frac{2^{1-N(N-1)}}{N!(2\pi)^{N-1}\sin^{N^2}\alpha} > 0. \tag{10.3}$$

This is the coefficient computed in Section 4, including the determinant sign, all one-body factors, $1 + Z$, $1/N!$, and the normalized constrained measure. All parameter derivatives of A preserve the displayed Vandermonde zeros, since the real angular coordinates are independent of s.

Lemma 10.3. *Put*

$$a_0 = \frac{N^2-1}{2}, \qquad d_\alpha = \cot\alpha, \qquad \mathcal{C}_N = \frac{1}{\sqrt{N}} \int_{\substack{\sum_i \omega_i = 0 \\ |\omega|=1}} \Delta(\omega)^2\, d\sigma(\omega) > 0.$$

Then, with $\tau_N = -\sum_{i<N} \tau_i$,

$$\begin{aligned} J_\alpha &= \int_{\mathbb{R}^{N-1}} \frac{\Delta(\tau)^2}{(2N - id_\alpha \sum_i \tau_i^2)^{k+1}}\, d\tau_1 \cdots d\tau_{N-1} \\ &= \frac{\mathcal{C}_N}{2}(2N)^{-1/2}(-id_\alpha)^{-a_0} B(a_0, 1/2) \neq 0. \end{aligned} \tag{10.4}$$

Here $\arg(-id_\alpha) = -\pi/2$.

Proof. The coordinate measure on the zero-sum hyperplane is Euclidean measure divided by $\sqrt{N}$. Polar coordinates give radial numerator R^{N^2-2}. Substitution $u = R^2$ gives the beta integral for a positive real quadratic coefficient; continue that coefficient through the fourth quadrant to $-id_\alpha$. The uniform majorant is a constant times $R^{N^2-2}(1+R^2)^{-N^2/2}$, whose tail is R^{-2}. The angular constant is positive because a zero-sum vector with distinct coordinates has an open spherical neighborhood where $\Delta^2 > 0$. Every remaining factor is nonzero. □

Theorem 10.4. *At the selected prime root,*

$$C_N^{(k)}(s_\varepsilon) = \Lambda_\alpha \varepsilon^{-1/2} + o(\varepsilon^{-1/2}), \qquad \Lambda_\alpha = 2A_\alpha(-1)^k k! D_s^k J_\alpha \neq 0.$$

For $j < k$, $C_N^{(j)}(s_\varepsilon)$ remains bounded. Every even $n < N$ has bounded derivatives of every fixed order along this approach.

Proof. By Lemma 10.1, $C_N - 2I_N$ has bounded fixed derivatives. Differentiate (10.2) k times. Every term with denominator power at most k is uniformly integrable, since its radial exponent at $\varepsilon = 0$ is at least

$$N^2 - 2 - 2k = 0.$$

The only term of possible size $\varepsilon^{-1/2}$ is

$$\int \frac{\chi(t)\Delta(t)^2(-1)^k k!(\partial_s q(t, s_\varepsilon))^k A(t, s_\varepsilon)}{q(t, s_\varepsilon)^{k+1}}\, dt.$$

Set $t = \sqrt{\varepsilon}\,\tau$. Its total scale exponent is

$$\frac{N(N-1)}{2} + \frac{N-1}{2} - (k+1) = -\frac{1}{2}.$$

Lemma 10.2 gives the integrable majorant $C\Delta(\tau)^2/(1+|\tau|^2)^{k+1}$. Dominated convergence, (10.1), and (10.3) give $A_\alpha(-1)^k k! D_s^k J_\alpha$ for the lower chart. The factor two in Λ_α counts the two reflected centers; the separate factor two in A_α comes from $1 + Z = 2$. For $j < k$ every denominator power is at most k, so the same radial integrability argument and the complementary lemma prove boundedness. □

The radial phase is also explicit: $k = 2p^2 - 1 \equiv 1 \pmod 4$ implies that $\zeta^k \Lambda_\alpha$ is a positive real multiple of $e^{-i\pi/4}$.

11 Noncancellation and the natural boundary

Let

$$U(s) = \sum_{\substack{N \geq 2 \\ N \text{ even}}} C_N(s) = \mathcal{X}(s)/\mathcal{M}^2(s).$$

The normally convergent exterior series may be differentiated term by term. At a selected prime root, the finitely many lower orders have bounded kth derivatives by Lemma 10.1. Theorem 10.4 gives the first singular term, and Theorem 9.1 controls the complete higher tail. Consequently

$$U^{(k)}(s_\varepsilon) = \Lambda_\alpha \varepsilon^{-1/2} + o(\varepsilon^{-1/2}), \qquad \Lambda_\alpha \neq 0. \tag{11.1}$$

Thus U cannot extend holomorphically through ζ. Since $\mathcal{M}^2$ is analytic and nonzero at this nonaxis point, the exact identity $\mathcal{X} = \mathcal{M}^2 U$ transfers noncontinuation to the full connected bulk function.

To obtain the physical leading coefficient, set $g(\varepsilon) = U((1+\varepsilon)\zeta)$. This function has $g^{(k)}(\varepsilon) = O(\varepsilon^{-1/2})$. Integrability gives a finite limit of $g^{(k-1)}$ at zero; successive integration gives finite limits of all lower derivatives. Leibniz's rule therefore yields

$$\mathcal{X}^{(k)}(s_\varepsilon) = \mathcal{M}^2(\zeta)\Lambda_\alpha \varepsilon^{-1/2} + o(\varepsilon^{-1/2}).$$

The factor $\beta_{\text{phys}}(s) = \operatorname{arsinh}(s)/(2J)$ is locally analytic and nonzero at every selected point on the branch continued from the physical interval. Multiplication by it gives the same obstruction for χ.

The selected prime roots are dense in the open first quadrant. Evenness and conjugation give a dense set of obstructions in all four quadrants. Every neighborhood of a point on the unit circle, including an axis point, contains such an obstruction. Hence no point of the circle admits a holomorphic continuation of the exterior germ. This proves Theorem 1.1.

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